Deliverable D - Technical Writeup

Team: Closed-Loop Underwriting

> *Numbers below are recomputed from the raw CSVs and from a proxy scorecard (`scripts/run_scorecard.py`) measured where ground truth exists. The TRUE competition score is not locally computable (hidden test labels + hidden per-term normalization), so every figure we quote is a **proxy** measured on the out-of-time **validation funded** holdout (2,551 loans applied after the train window), aggregated with the brief's published weights (p.14).*

1. Problem framing & assumptions violated

This is a **selective-labels** underwriting problem, not a clean classification task. Outcomes (`default_flag`, timing, recovery) exist only on funded+matured loans - 51,722 of 85,340 train rows; **0 of 8,817 test rows**. Three standard assumptions break, and each changed our design:

- **Positivity fails globally.** The legacy funding rule is *deterministic*:

`prior_decision == 1 iff prior_underwriter_score >= ~0.273`, with zero mismatches and a clean gap (max declined 0.27297 < min approved 0.27301). There are no funded look-alikes for declined applicants, so IPW / doubly-robust estimators are **not identified**. We therefore fit the outcome model unweighted and treat declines by **bounding / abstention**, never silent extrapolation. We also **exclude** `prior_underwriter_score` **and its consequences** (`prior_decision`, `prior_approved_amount`) from the PD features to avoid encoding the selection.

- **The payoff is sharply asymmetric and time-dependent.** A repaid loan nets

`0.0875 * R` (3% fee + 35% * 60/365 interest); a default destroys most of `R` net of recovery. Crucially, default *timing* is bimodal: in-term defaults span days 3-60, then a spike at exactly day 90 (the observation-window close) with **zero** mass in (60, 90). The day-90 spike recovers ~=0 (empirical mean recovery fraction **0.0001** vs **0.118** in-term) - these are near-total losses, not near-payers.

- **Missingness is structural.** The six bank-feed columns are null *iff*

`has_linked_bank_feed == False`. We preserve NaN (HistGB splits on it) and add explicit `*_was_null` indicators rather than imputing.

2. Methodology

One **weekly competing-risks discrete-time hazard** (default vs payoff) is the spine for all four deliverables. A single 3-class `HistGradientBoostingClassifier` on a person-period expansion yields per-week hazards `h_d, h_p`; the competing-risks recursion gives `CIF_d(t)` and lifetime PD. Deliverables read off one fit. Stack is **sklearn-only** (HistGB + isotonic); features are all-float with NaN preserved.

- **Deliverable A - timing-integrated NPV policy.** We replaced a flat

break-even rule (`approve if PD < ~8.9%`) with an **expected-NPV rule that integrates over default timing**: `E[NPV_i] = sum_t P(default in week t) * NPV(t) + P(repay) * 0.0875R`, where `NPV(t)` credits the daily ACH draws a *late* in-term defaulter pays before defaulting, and day-90-window defaults are charged as total losses (no term draws). Approve iff `E[NPV] > 0`. A flat PD threshold over-declines profitable late-defaulters; the timing rule recovers them. **Realized backtest (high compute):** timing earns **$603,817** (approving 59%) vs the conservative flat rule's **$512,660** (approving 27%) on the 2,551-loan funded holdout (**+$91,157**), against -$2.56M for approve-all/legacy and a $4.75M hindsight ceiling. The extra approvals timing wins back are exactly the late-in-term defaulters whose pre-default ACH draws keep their E[NPV] positive.

- **Deliverable B - cohort trajectory.** Approved-cohort `CIF_d` averaged per

(cohort_week, age), monotone by construction. The hazard mildly under-predicts out-of-time (lifetime ratio ~=1.12), so we apply a **single-parameter OOT recalibration** fit on the validation funded subset: weighted CDR MAE drops from **0.0207 -> 0.0150** (naive train-marginal baseline 0.0259).

- **Deliverable C - counterfactuals.** `do(feature=value)` via the causal-graph

surgical intervention (see Section 3).

All scaling is governed by one `--compute {low,med,high}` flag (ensemble size, conformal splits, harness grid) and is deterministic per (seed, budget).

3. Causal reasoning & counterfactual methodology

Deliverable C asks for an **interventional** `P(default | do(X=x))`, not the observational `P(default | X=x)`. Our submitted estimator applies `do(X=x)` to the applicant's raw row, **propagates the deterministic descendants** via the DAG (`dag.propagate_intervention`: engineered ratios recomputed; toggling `has_linked_bank_feed` rewrites the whole bank-feed block + its missingness flags), then reads lifetime PD off the hazard ensemble. For features with SCM descendants this is **standardization / g-computation in spirit**; for the rest it is a disclosed observational perturbation - we say so plainly rather than overclaiming a full SCM on real data where positivity fails.

Because real data **cannot** validate interventional accuracy (we never observe counterfactuals), we built a **fidelity-gated synthetic validation harness** (`closed-loop-default-detection`): a structural causal model tuned to match the real marginals (fidelity gate green), with a true `do()` oracle. There we grade a deployable **g-computation estimator** (fit on approved rows only, no SCM coefficients) against **naive conditioning**. Certified across a **5-seed sweep** (seeds 7/13/42/101/2026, 900 queries each): on the **strong-propagation** slice at moderate selection severity (0.4), g-computation MAE is **0.0797 +/- 0.0135 vs naive 0.0988 +/- 0.0154** - gap **+0.0191 +/- 0.0046, positive on 5/5 seeds with no sign flips**, a **~19% relative reduction** in interventional error exactly where naive conditioning is structurally wrong (it ignores that intervening on a parent moves its children).

The harness also measures where this stops working. At **full selection severity (1.0)** the strong-propagation gap collapses to **+0.0021 +/- 0.0022 and flips sign on one seed** - a statistically zero advantage, and we claim none. This is the same limit that breaks the IPW selective-labels frontier between severity 0.4 and 0.6 (Section 4): estimators fit on approved rows whose conditionals are distorted by selection on an *unobserved* confounder. Backdoor adjustment cannot fix unobserved confounding; IPW cannot fix broken positivity - **one structural mechanism bounds both the observational reweighting fix and the causal estimator**, measured independently in the same synthetic world. **Regulator defense:** drivers are reported as interventional effects with the propagation made explicit and non-intervenable identity features (sector, vintage) refused - not raw observational correlations - and the validation states both halves: the direction of the method is certified with error bars across seeds, *and* the regime where it stops working is measured, not assumed away.

4. Calibration & uncertainty quantification

The scoreboard rewards 90% intervals that **cover ~=0.90 at minimum width**, and decisions flip in a narrow band near break-even, so calibration is first-class.

- **A (PD).** Raw ensemble percentile bands measure model *disagreement*, not

predictive uncertainty for a binary outcome, and they **under-cover** (binned coverage 0.70 at high compute). We use a **split-conformal** half-width: bin by predicted PD, take the 0.95-quantile of per-bin `|empirical_rate - predicted|`, fit on the validation funded subset. Conformity is measured on the **raw** point - a *fitted* recalibrator (isotonic) shrinks in-fold error and under-covers out-of-fold (we verified raw -> 0.875 vs isotonic -> 0.53 held-out coverage). Result on the holdout (evaluated by repeated 50/50 split-conformal, no circularity): **coverage 0.900 at width 0.133** (vs raw 0.70 / 0.079).

- **B (CDR).** Ensemble percentile bands across members, monotone per cohort.
- **C (PD_cf).** Ensemble percentile bands across members; population fallback for

unscoreable queries so the file is never null.

Declined-subpopulation coverage is the part real data can never check; the harness measures it against SCM truth - and the selective-labels loop now runs **on the SCM itself**, the same synthetic world as the Section 3 counterfactual results. IPW holds declined-cohort ECE at **0.025 / 0.037 / 0.087**

for severity 0 / 0.2 / 0.4 (pass, seed 42), then fails at 0.6 (**0.250**) - an honest operating frontier at severity 0.4, the same boundary where the g-computation advantage collapses (Section 3), measured in one world rather than two.

5. Limitations & what we'd do differently

- **Proxy != truth.** All gains are proxy-measured on one OOT holdout; the hidden normalization could reweight terms. We mitigated by recomputing every number from raw CSVs and cross-fitting calibration to avoid optimism.
- **The day-90 spike convention is load-bearing** for P&L (30%). We charge it as a total loss (no term draws) on strong empirical grounds (recovery ~=0.0001), but a different draw-accounting convention would move the P&L level.
- **C is g-computation-in-spirit, not a fitted SCM on real data** - positivity failure makes a full real-data SCM unidentified. The harness certifies direction across 5 seeds **only inside the severity <= 0.4 frontier**; at full selection severity the advantage is statistically zero (sign flip on one seed), and the MAE gain trades a small but consistent increase in negative bias (g-comp bias more negative than naive on 5/5 seeds) - measured limits, not the exact real-data magnitudes.
- **With another day:** fold the conformal recalibration into the *decision* (not just reported PD) to close the gap to the $694K ranking optimum; per-cohort (not global) OOT recalibration for B; bootstrap bands for C scaled by compute.